

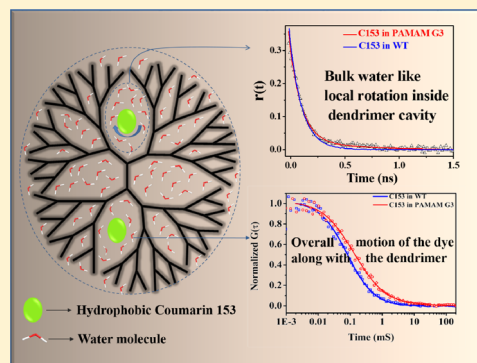
Encapsulation and Residency of a Hydrophobic Dye within the Water-Filled Interior of a PAMAM Dendrimer Molecule

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Supporting Information

ABSTRACT: Tightly confined water within a small droplet behaves differently from bulk water. This notion is obtained on the basis of several reports showing unusual behaviors of water droplet residing at the core of a reverse micelle. In this study, we have shown a well-known hydrophobic dye, coumarin 153 (C153), which prefers to reside at the water-rich region inside the dendrimer molecule. Optical density (OD) measurement at the absorption peak of C153 shows that it is almost insoluble in bulk water but highly soluble in aqueous dendrimer solution. The OD of C153 increases several times in the latter case as compared to that in the former. We found the most interesting observation when we compared the data from fluorescence correlation spectroscopy (FCS) with the fluorescence anisotropy decay of C153 in aqueous dendrimer solution. The FCS measurement reveals a much slower translational diffusion time (τ_D) of C153 attached to a dendrimer molecule as compared to that of free C153 in bulk water in the absence of dendrimer. The slower τ_D in the former case is commensurate with the size of the dendrimer molecule. This is possible only when C153 is encapsulated by the dendrimer molecule. In contrast to the FCS study, the fluorescence anisotropy decay of C153 in water remains largely invariant after addition of the dendrimer. This can happen if a bulk-water-like environment at the C153 surroundings is preserved within the C153–dendrimer complex. This supports our intuitive expectation that C153 resides within the water-rich peripheral cavities of the dendrimer molecule. A more expected binding of C153 to the hydrophobic core of dendrimer may not be possible here because of an inadequate size of the dendrimer core.



1. INTRODUCTION

Highly symmetric and hyperbranched dendrimers (Schemes S1 and S2) have attracted considerable attention in various research fields because of their widespread applications ranging from nanoparticle synthesis to drug delivery and most recently in gene therapy.^{1–9} Different-generation dendrimers with various structural scaffolds have been used in recent reports for various biomedical applications, quantum dot synthesis, optical sensing, mimicking of globular proteins, and optical imaging of cells.^{2–7,10,11} Soft molecular interactions (hydrophobic/hydrophilic, electrostatic and hydrogen bond interactions) between guest molecule and dendrimer lead to the formation of a stable supramolecular dendritic complex.^{2,4} There have been a few number of reports in the literature that show molecular-level insight to a host–guest interaction in a dendrimeric complex.^{12–16} Here, we studied the interactions between hydrophobic and hydrophilic coumarin dyes with amine-terminated first (G1)- and third (G3)-generation poly(amidoamine) (PAMAM) dendrimers (Schemes S1 and S2). Amine-terminated dendrimers bear two different types of nitrogen centers: primary amines at the periphery and tertiary amines at the interior. These two amine centers behave differently as the medium pH is changed.^{17,18} At neutral pH (~ 7), primary amine ($pK_a \sim 9.2$)¹⁷ groups at the exterior of dendrimer remain completely protonated, whereas tertiary

amines ($pK_a \sim 6.3$) at the interior of dendrimer remain partially protonated within the aqueous solution (Scheme S2). Stepwise protonation of the primary amines first and tertiary amines next with lowering of the pH of the dendrimer solution has a close relationship with the pH-dependent fluorescence intensity change of the dendrimer molecules.¹⁸ For a G4-amine-terminated dendrimer, Imae and her co-worker observed an insignificant change of fluorescence intensity when pH was lowered from ~ 11 to ~ 6 .¹⁹ This was followed by a dramatic enhancement of the emission intensity (without shifting the emission peak position) on further lowering the pH of the medium. They obtained the maximum intensity at pH ~ 2.5 , and no change of intensity was noted thereafter. They attributed this observation to a number of facts: the strengthening of hydrogen bonds at low pH; the formation of a rigid structure of dendrimer in acidic medium due to the electrostatic repulsions among the protonated tertiary amine centers; and the formation of a new highly fluorescent molecule in low-pH medium.¹⁹ Their explanation tacitly assumes that the fluorescent center is planted deep inside the dendrimer molecule as it is mostly influenced only by the protonation of

Received: October 8, 2016

Revised: January 4, 2017

Published: February 6, 2017



the interior amines (at low pH). The intrinsic fluorescence of dendrimer is also modified by several other factors, such as the formation of aggregates among the dendrimer molecules, the nature of the terminal group (amine, hydroxy, or carboxylate) present at the exterior surface of dendrimer, and the nature of the explicit solvent, where dendrimer is dissolved.^{20,21} Recently, Prasad and his co-worker have observed that dendritic aggregations, formed by the electrostatic interactions among the oppositely charged dendrimers, mostly follow a fascinating dendritic pattern.²¹ Within these patterns, nonradioactive pathways of exciton decay are significantly reduced, which results in a considerable appreciation to the intrinsic fluorescence from dendrimer molecules.²¹ The radius of gyration (R_g) of dendrimers, in contrast to the dendrimer fluorescence (which is sensitive to the medium pH), does not change much (<4%) with the medium pH (approximately 4–10). Previous studies of small-angle neutron scattering (SANS) on G8 and G4 PAMAM dendrimers reveal only ~2 and ~4% increase of R_g , respectively, as the pH of the medium is changed from ~10.1 to ~4.7.^{22,23} Findings of these SANS studies are highly congruent with those of a recent work on atomistic molecular dynamics (MD) simulations by Goddard III and his co-workers.²⁴ They observed a subtle swelling (<4%) of G4 PAMAM dendrimer in aqueous solution as the pH of the solution was reduced.²⁴ They further plotted (solvent accessible surface area)^{1/2} and (solvent excluded volume)^{1/3} of dendrimers as a function of the size of probe molecules. From these plots, they concluded that probe molecules with size >1.4 nm cannot be placed inside a dendrimer molecule.²⁴

In the present study, we show an extremely hydrophobic dye, coumarin 153 (C153), which strongly interacts with dendrimers to form a stable dendritic host–guest complex at all pHs (3–10). However, a similar-size hydrophilic dye, C343, does not interact equally with dendrimer within the same pH range. The maximum interaction between C343 and dendrimer was observed around pH ~9; at this pH, the dendrimer exterior bears positive charges, whereas C343 prevails in its deprotonated form (negatively charged). At a very high (>11) or a very low (<2) pH, C343 hardly interacts with the dendrimer molecules. In the C153–dendrimer complex, quasi-confined water molecules inside the dendrimer act as a better solvator than bulk water for C153.²⁵ However, for C343, the driving force for complex formation is the strong electrostatic interaction, which becomes feasible around pH ~9. This work demonstrates that a judicious selection of medium pH will help control binding of dendrimer with drug molecules bearing a similar functional group to C343. Apart from noncovalent interactions (hydrophobic/hydrophilic, hydrogen bond, coulombic interactions, etc.), the confinement effect on the trapped water inside dendrimer plays a crucial role in dissolving the hydrophobic molecule at the dendrimer interior.²⁵

2. EXPERIMENTAL SECTION

2.1. Materials. Generation 1 and generation 3 of PAMAM dendrimer (20 wt % in methanol), C153, coumarin 343 (C343), methanol, hydrochloric acid, and sodium hydroxide pellets were purchased from Sigma-Aldrich. Milli-Q water (Merck Millipore) was used for the preparation of aqueous dendrimer solutions.

2.2. Instrumental Techniques. Steady-state absorption and fluorescence spectra were recorded on PerkinElmer Lambda 750 spectrometer and Cary Eclipse fluorescence spectrometer, respectively. Fluorescence lifetime data were

collected using a time-correlated single photon counting (TCSPC) setup obtained from Edinburgh Instruments. Pico-second-pulsed diode lasers of wavelengths 375, 405, and 445 nm were used to excite the dendrimer-G1, dendrimer-G3, and coumarin dyes, respectively. Instrument response functions (IRFs) of the diode lasers were found to be $\sim 80 \pm 10$ ps. Lifetime decays were analyzed by F900 software provided with the TCSPC instrument. For time-resolved rotational anisotropy measurements, we used the same TCSPC setup. During the anisotropy measurements, we periodically altered the direction of the emission polarizer from parallel to perpendicular and vice versa with respect to the excitation polarizer at regular time intervals. The rotational correlation function was calculated by the following equation

$$r(t) = \frac{I_{\parallel}(t) - GI_{\perp}(t)}{I_{\parallel}(t) + 2I_{\perp}(t)} \quad (1)$$

where $I_{\parallel}$ and $I_{\perp}$ are the fluorescence intensities at the parallel and perpendicular directions of the polarizer, respectively, and G is the instrumental correction factor. The G factor was obtained by tail-matching method using a dye, which rotates freely within the conventional solvents (i.e., C153 in methanol). Fluorescence correlation spectroscopy (FCS) of C153 was studied using a confocal microscope (MicroTime 200) from PicoQuant. Pulsed diode laser of wavelength 405 nm was used to excite the C153 samples placed on a coverslip. The laser power and pinhole diameter were $\sim 30 \mu\text{W}$ and $\sim 50 \mu\text{m}$, respectively, throughout the course of the experiment. The excitation light was focused on the sample through a water immersion objective (60 $\times$, 1.2 NA). The transverse radius ($\omega_{xy} \sim 310$ nm) of the excitation volume was calibrated using a sample with a known diffusion coefficient (rhodamine 6G in water; $D_t = 426 \mu\text{m}^2/\text{S}$).²⁶ The hydrodynamic radius of a particle diffusing through the confocal volume was obtained from the Stokes–Einstein equation (eq 2), using the D_t value from FCS measurement²⁷

$$D_t = \frac{k_B T}{6\pi\eta r_h} \quad (2)$$

Dynamics light scattering (DLS) measurements for the dendrimer-G3 samples at different pHs were performed using a Malvern Zetasizer nano series instrument (version 7.03). All experiments were carried out at a constant temperature (25 °C) using He–Ne laser at a scattering angle of 173°.

2.3. Sample Preparation and Evidence for Host–Guest Complex Formation. Among the several methods, we attempted dissolving of C153 in aqueous dendrimer solution and found that the solvent evaporation method is the most efficient one, in which the host molecules (dendrimer) along with the hydrophobic dyes C153 (guest) are dissolved in a common solvent (methanol), where both are equally soluble, and the solvent is slowly evaporated to produce a thin film at the wall of the round-bottom flask. This film is hydrated with excess water, and thereby the encapsulated guest molecules within the host are solubilized in milli-Q water.^{28,29} The partitioning of C153 in aqueous phase in the presence of dendrimer is confirmed by checking the UV–vis spectrum, where a strong characteristic absorption band of C153 is observed. As C153 is almost insoluble in water, a maximum OD of $\sim <0.02$ can be found from a C153-saturated water solution. However, the OD of C153 increases several times (>14 times) within the aqueous ~ 0.2 mM dendrimer solution (Figure 1).

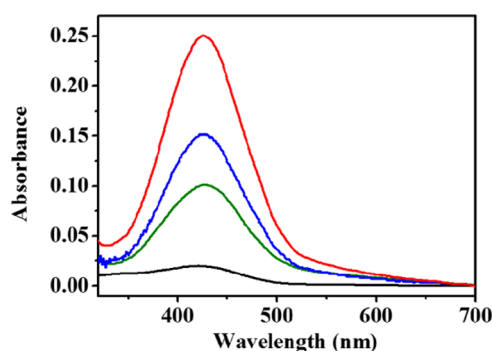


Figure 1. Absorption spectra of C153-saturated water in the absence (black curve) and the presence of ~ 0.05 mM (green curve), 0.1 mM (blue curve), and 0.2 mM (red curve) dendrimer-G3 (pH ~ 9.6). In the presence of dendrimer, the solubility of C153 increases dramatically because of the formation of a stable water-soluble dendrimer–C153 complex. Emission spectra of the C153–dendrimer complex can be found in Figure S2.

C153-saturated water is apparently transparent, and almost no emission from C153 can be found when excited with UV light. However, in the presence of ~ 0.2 mM dendrimer, because of the higher solubility of C153, the resultant solution (C153-saturated aqueous dendrimer solution) turns yellow. A strong C153 fluorescence is observed from this solution when excited with a UV light (Figure S1). This fact confirms that the dendrimer solubilizes the hydrophobic dye, C153, in water through the formation of a stable water-soluble dendrimer–C153 complex.

3. RESULTS AND DISCUSSION

3.1. Study of Red-Edge Excitation Shift (REES) Using Intrinsic Fluorescence of Dendrimer and Molecular Recognition Study of the C153–Dendrimer Complex.

PAMAM dendrimers are intrinsically fluorescent in low-pH solvents.¹⁹ In the present work, we extensively studied the intrinsic fluorescence from generation 1 (G1) and generation 3 (G3) of the PAMAM dendrimer only at natural pHs. The pH-dependent fluorescence properties of the same dendrimer but different generations (G2 and G4) are already reported in a seminal work by Imae and her co-worker.¹⁹ Imae et al. and Crooks et al. noted few remarkable spectral properties of dendrimer molecules, which are worth mentioning here before going to our REES experiment.^{18,19} (1) Most of the PAMAM dendrimers exhibit an absorption peak ($\lambda_{\text{abs}}^{\text{max}}$) within the range of 280–290 nm (Figure 2; black curve), and excitation at the absorption peak of dendrimers produces a broad fluorescence spectrum with weak intensity.¹⁶ (2) Excitation spectra, as obtained from the plot of emission intensity at emission peak wavelength against excitation wavelength, were found incapable of replicating the actual absorption spectrum. Interestingly, these excitation spectra (peaked within approximately 330–400 nm) are well structured and highly red-shifted compared with their corresponding absorption spectra (Figure 2; blue and red dotted curves). (3) Excitation at the peak of the excitation spectrum provides a highly intense and narrower emission spectrum compared to those of the emission spectrum obtained through the excitation at absorption peak (Figure 2).¹⁶

In REES experiment, we therefore studied the emission spectra of dendrimers by exciting within the ranges of 310–405 and 335–430 nm for dendrimer-G1 and dendrimer-G3,

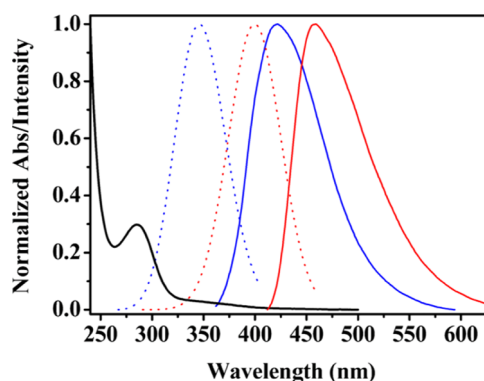


Figure 2. Absorption spectrum of dendrimer-G3 (black) in water. Excitation (dotted lines) and emission (solid lines) spectra of dendrimer-G1 (blue) and dendrimer-G3 (red) in water. Absorption spectra of G3 and G1 are similar. Only G3 absorption spectrum is provided in the figure for clarity, and pH of these aqueous dendrimer solutions was found to be ~ 9.6 .

respectively, with an ~ 5 nm interval between two successive excitation wavelengths (exciting around the excitation spectra of G1 and G3 dendrimers). The emission peak wavelengths from both the generations are shifted toward red wavelengths when excitation energies are gradually moved from the high-energy side to the low-energy side. This emission peak shifting continues without showing any tendency of convergence, neither at the low-energy side nor at the high-energy side of excitation wavelengths. (Figures 3 and S3). This phenomenon is called REES. REES provides an idea of static inhomogeneity within the medium in a quantitative sense.^{30,31} Excitation at

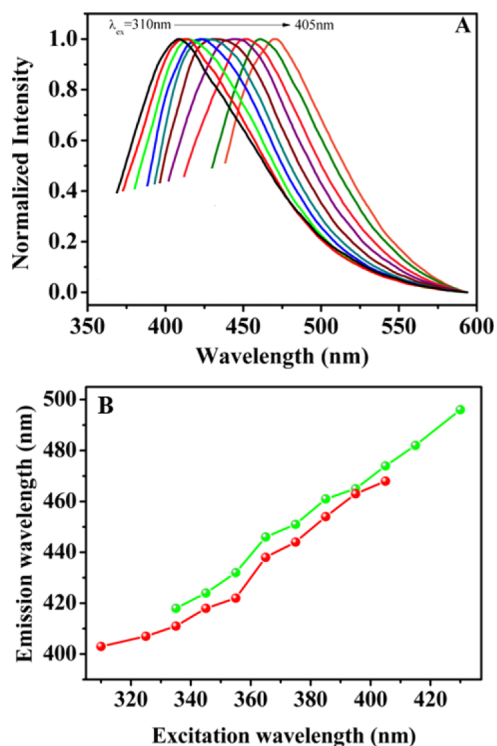


Figure 3. (A) Emission spectra of aqueous dendrimer-G1 (pH ~ 9.6) at different excitation wavelengths (for dendrimer-G3, see Figure S3). (B) Plots of emission wavelengths against excitation wavelength for dendrimer-G1 (red) and dendrimer-G3 (green) in aqueous solution (pH ~ 9.6).

extreme blue wavelengths (<300 and <320 nm for dendrimer-G1 and dendrimer-G3, respectively) leads to a broad and weak emission spectrum with one shoulder-like peak from both the generations, which restricted our excitation wavelengths to ~310 nm (for dendrimer-G1) and 335 nm (for dendrimer-G3) at higher-energy side of excitation.¹⁷ The amount of REES is more (~78 nm) in G3-dendrimer compared to that (~65 nm) in dendrimer-G1 (Figure 3B, Table S1).

The actual source of inhomogeneity (as indicated by significant amount of REES) associated with monodispersed dendrimer molecules is not very clear. However, several groups have reported that the nonideality of the synthesis methods causes implantation of random defect sites within the dendrimer molecules.^{32–34} These defect sites, in turn, can produce intrinsic inhomogeneity of dendrimer molecules.^{32–34} A defect site may arise from several components: (1) skeleton defects with missing fragments or the formation of unwanted bonds connecting the branches; (2) the presence of “stray fragments” inside a dendrimer molecule; and (3) the existence of low-generation dendrimers obtained from the presence of excess core fragments at the time of growth to a higher-generation target.³⁴ A higher amount of REES (~78 nm) in dendrimer-G3 compared to that (~65 nm) in dendrimer-G1 is well under our intuitive expectation; a higher-generation dendrimer with a larger number of defect sites will produce a more inhomogeneous environment compared to that in a lower-generation dendrimer with a lower number of defect sites. Additionally, lower-generation dendrimers bear relatively open structures as compared to those adopted by higher-generation dendrimers. As the generation of dendrimer grows, repulsion among the terminal groups causes significant back-folding of the peripheral subgroups. This back-folding leads to a densely packed inhomogeneous globular-protein-like structure within the higher-generation dendrimer molecules.^{19,35–37} REES within a globular-like protein is a well-known phenomenon reported by several groups.³¹ Hence, the globular-like structure of the higher-generation dendrimer may be the reason for a larger REES from dendrimer-G3 compared to that observed for dendrimer-G1.

In the next step, we added a water-insoluble hydrophobic dye (C153; ~6 μ M) to the aqueous G3-dendrimer solution (~0.2 mM) through the solvent evaporation method. It is worth mentioning here that a high solubility of C153 within the aqueous dendrimer solution, which is in conjunction with a higher quantum yield of C153 compared to that of dendrimer in aqueous solution, overshadows the weak dendrimer emission by the strong C153 fluorescence of C153 in aqueous dendrimer solution (Figure S2). Moreover, the location of the emission peak of dendrimer (in water) is different from that of C153 in aqueous dendrimer solution. These factors enabled us to successfully monitor C153 fluorescence from the C153–dendrimer complex formed in aqueous solution. Interestingly, here we observed no REES from C153 fluorescence when attached to a dendrimer molecule. This may be due to the local homogeneous environment of C153 within the complex. The emission peak position (~552 nm) and QY (~0.08) of C153 in the dendritic complex are close to the corresponding figures (~549 nm and ~0.12) of C153 in bulk water.³⁸ This fact ascertains that C153 in the dendritic complex is located within the hydrophilic region, which is either at the exterior (dendrimer periphery) or within the internal cavities of the dendrimer. In both the places, C153 can get a waterlike environment. Solvation dynamics study utilizing the fluores-

cence of C153 (attached to dendrimer) can help us reveal the nature of immediate environment of C153, within its complex with the dendrimer molecule. If C153 resides in a waterlike environment within the complex, then we can expect an ultrafast (few picoseconds) solvation dynamics (therefore, the ultrafast rise components at red emission wavelengths of C153); however, it cannot be detected by our TCSPC setup (IRF ~ 80 ps).³⁹ On the other hand, if C153 binds to a hydrophobic site of dendrimer, the rise components at the red emission wavelengths will be slower by several orders of magnitudes, which can be easily detected by our setup.^{39–44} As expected, we did not find any detectable rise component in the lifetime decay curves of C153 (in aqueous dendrimer solution) at red emission wavelengths using a TCSPC setup (Figure S4). Therefore, this observation also indicates that C153 resides in a hydrophilic environment within the complex.

3.2. Study of C153 Fluorescence from the Dendrimer–C153 Complex as a Function of Medium pH.

Intrinsic fluorescence of dendrimers is greatly enhanced with lowering medium pH, especially when tertiary amine centers are protonated.¹⁹ Under this condition, the amine centers are unable to quench the excited-state fluorophore units of dendrimers through electron transfer.⁴⁵ In this work, we studied the effect of medium pH on the fluorescence property of C153 attached to a dendrimer molecule. Similarly to the intrinsic fluorescence of dendrimer, which increases significantly with lowering of the medium pH, the fluorescence intensity of C153 attached to dendrimer increases as the medium pH is lowered. However, in the latter case, the intensity change is not very significant. In an aqueous dendrimer solution (pH ~ 9.6), only primary amine centers at the dendrimer exterior remain protonated. QY of C153 in this solution is ~0.075. However, when we lowered the pH to ~2 by adding HCl, both the primary and tertiary amine centers became protonated.¹³ Within this low-pH dendrimer solution, QY of C153 was found to be ~0.11, which is very close to that of C153 (~0.12) in bulk water (Figure 4A).³⁸ In the former solution (pH ~9.6), C153 fluorescence is quenched by photoinduced electron transfer (PET) from nonprotonated tertiary amine centers, located in the dendrimer interior, to the excited C153 molecule.¹³ When pH was decreased to ~2 in the latter solution, all amine centers became protonated, and quenching due to PET from amine to excited-state C153 was not possible.¹³ Bulk-water-like QY of C153 from a low-pH dendrimer solution is a clear indication of C153 residing within the waterlike environment inside the dendrimer molecule. The fluorescence enhancement of C153 caused by the protonation of interior tertiary amine centers of dendrimer implies that C153 is also located at the interior of a dendrimer molecule, very close to the tertiary amine centers. Imae and her co-worker have recently reported a similar observation that the fluorescence intensity of pyrene at 446 nm decreases when it binds to the interior of a poly(propyleneimine)(G3)–core:PAMAM(G3)–shell dendrimer.¹³ According to them, fluorescence quenching of pyrene here is due to PET from tertiary amine centers of dendrimer to the excited pyrene molecule.¹³

Before arriving to our conclusion that protonated tertiary amine is unable to take part in PET at low pH, which leads to C153 fluorescence unquenched, direct evidence showing the protonation of tertiary amine centers of dendrimer at low pH (<5.5) is highly desirable. For this, we performed a quantitative pH titration of aqueous dendrimer-G3 solution by slow

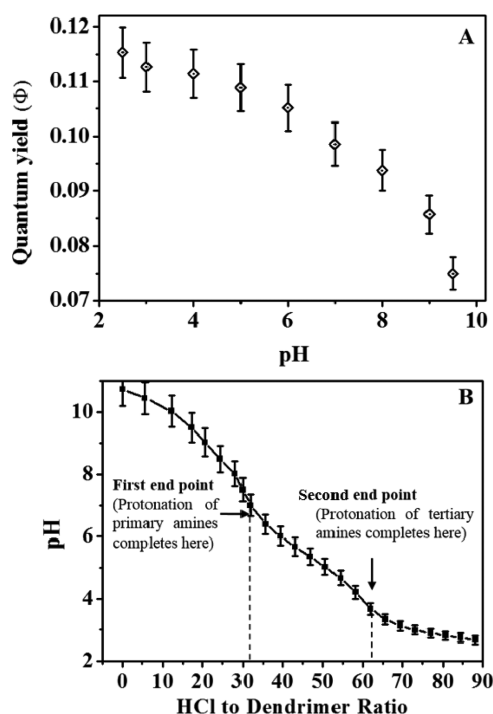


Figure 4. (A) Plot of fluorescence quantum yield (Φ) of C153 in aqueous dendrimer-G3 solution against medium pH. (B) Plot of pH of aqueous dendrimer-G3 solution against HCl/dendrimer mole ratio. Two end points were detected during the titration because of the stepwise protonation of the primary amine centers (at HCl/dendrimer ~ 32) and then the tertiary amine centers (at HCl/dendrimer ~ 62). A good correlation of end points with the number of primary and tertiary amines present in dendrimer-G3 is observed. To obtain the entire pH range (approximately 11–2.5), we first increased the pH of the aqueous dendrimer solution from ~ 9.6 to ~ 11 by adding NaOH before commencing the titration.

addition of HCl (Figure 4B). One mole of dendrimer-G3 contains 32 moles of primary amine centers ($pK_a \sim 9.2$)¹⁷ and 30 moles of tertiary amine centers ($pK_a \sim 6.3$).³⁷ Therefore, the appearance of two end points in Figure 4B (pH vs HCl/dendrimer mole ratio curve) is due to the stepwise protonation of the primary amine centers (completed at ~ 32 molar ratio) and then the tertiary amine centers [completed at ~ 62 ($32 + 30$) mole ratio].

A careful examination of Figure 4A,B shows a good correlation; emission QY of C153 (in aqueous dendrimer solution) increases upon protonation of the tertiary amine centers (with decreasing the medium pH from ~ 8.5 to 4). This fact approves our belief that the fluorescence quenching of C153 at high pH is due to a PET from the tertiary amine center of dendrimer to excited C153. Therefore, C153 is located inside the dendrimer molecule. With regard to different binding sites within a dendrimer molecule, we found that C153 can possibly reach the hydrophobic core (EDA core).⁴² However, in a seminal work, Tomalia and his co-workers reported that the core size of an EDA core PAMAM dendrimer is too small to accommodate a fluorescent dye molecule.⁴⁶ Among other binding sites, C153 may enter into the water-rich cavities inside the dendrimer molecule. Goddard III et al. and several other authors showed that rose bengal, which is larger than C153, easily fits into the internal cavities of the dendrimer molecule.^{47–49}

To determine the exact location of C153 inside a dendrimer molecule, we studied the rotational anisotropy decay of C153 within the dendritic complex. If C153 is attached to the EDA core of the dendrimer, an excited-state depolarization of the C153 dipole can result through the rotation of the C153-attached dendrimer assembly as a whole, which would be much slower compared to the free rotation of C153 within the water-rich cavities of the dendrimer (if C153 resides within the cavity). Interestingly, we observed a fast single-exponential rotation of C153 at the dendrimer interior, which is highly congruent with the rotation of C153 in bulk water. No slow component of rotation was detected, which rules out the possibility of C153 getting trapped in the EDA core of the dendrimer. A detailed study of the rotational dynamics of C153 is discussed in the following section.

3.3. Rotational Anisotropy of Dendrimers Alone and C153 Attached to the Dendrimer. **3.3.1. Rotational Diffusion of Dendrimers Using Their Intrinsic Fluorescence.** Anisotropy decays using the intrinsic fluorescence of dendrimers (G1 and G3) in water well correlate to their corresponding hydrodynamic radii. The experimental anisotropy decay functions were conclusively fitted with the following biexponential fitting function

$$r(\tau) = r_0 \{a_1 \exp(-\tau/\tau_{\text{fast}}) + a_2 \exp(-\tau/\tau_{\text{slow}})\} \quad (3)$$

The initial anisotropy values ($r_0 \sim 0.3$, at $t = 0$; Figure 5A) from both the generations were found to be close to the limiting value of r_0 ($r_0 = 0.4$) for the rotation of a one-dimensional excited-state dipole.⁵⁰ The actual origin of the two rotational components (τ_{fast} and τ_{slow}) from both the generations of dendrimer is not very clear. To the best of our knowledge, this is the first report on the rotational anisotropy of PAMAM dendrimers utilizing their intrinsic fluorescence. Almost no information on the rotational dynamics of dendrimer molecules can be found from previous reports. The slow component (τ_{slow}) of the rotational decay is well correlated to the hydrodynamic radius (r_h) of the dendrimer molecule, as shown by the following Stokes–Einstein relation (eq 4)

$$\tau_{\text{slow}} = \frac{4\pi r_h^3 \eta}{3kT} \quad (4)$$

Therefore, the origin of τ_{slow} might be due to the overall rotation of the dendrimer molecule. In addition to the overall rotation, a local chain dynamics of nonrigid internal branches at the dendrimer interior may contribute to the depolarization of the intrinsic fluorescence of the dendrimer.⁵¹ This type of depolarization is more common for a dye molecule inside a micelle or noncovalently attached to a polymer chain.^{52,53} For simplicity, we assumed the internal movement of the fluorescent center of the dendrimer to be a wobbling type of motion, and its apex is fixed to the spherical surface of the dendrimer. This wobbling motion will produce a cone with a semicone angle of θ . The proposed model is essentially similar to the popular “wobbling-in-cone” model, which is frequently used in the spherical micellar systems. However, here, in contrast to the wobbling-in-cone model, we disregarded the translational diffusion of the fluorescent center. If $r(t)$ does not follow a biexponential decay, according to our proposed model

$$r(t) = r_0 [\beta \exp(-t/\tau_M) + (1 - \beta) \exp\{-t(1/\tau_M + 1/\tau_r)\}] \quad (5)$$

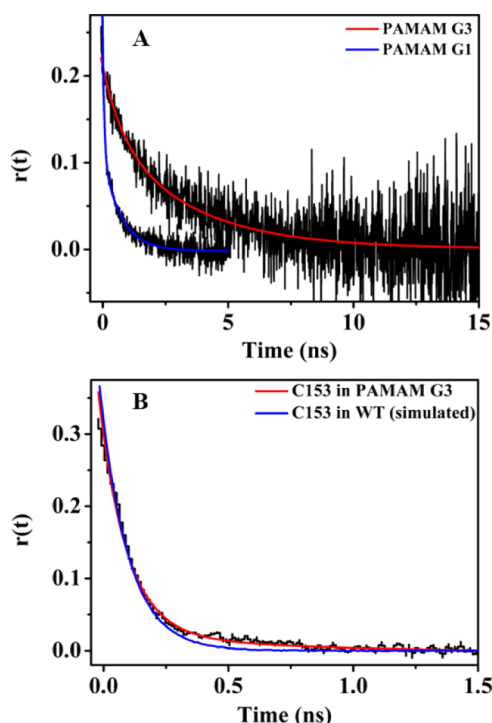


Figure 5. (A) Decay of rotational correlation function $r(t)$ with time for dendrimer-G1 and dendrimer-G3 in water. We utilized the intrinsic fluorescence of dendrimer for anisotropy measurement. Solid lines represent the fitted curves. (B) Rotational anisotropy decay curves for C153 in aqueous ~0.2 mM dendrimer-G3 solution (black curve). Red line represents the best fitting to the experimental decay curve. The blue line has been simulated using the reported rotational correlation time of C153 in bulk water from ref 57. The simulated curve is plotted with a viscosity ~ 1.15 times higher (to make a same viscosity with ~ 0.2 mM aqueous dendrimer-G3 solution) than that of neat water, to compare these two curves under isoviscous condition.

where r_0 is the initial anisotropy (at $t = 0$), β represents the fractional contribution to the overall anisotropy decay because of the rotation of the dendrimer as a whole, $\tau_M (= \tau_{\text{slow}})$ is the corresponding rotational correlation time, and τ_r is the time coefficient associated with the wobbling type of motion of the intrinsic fluorescent center of the dendrimer. Comparing eqs 3 and 5, we can correlate the experimental fitting parameters and the model parameters as follows

$$\tau_{\text{slow}} = \tau_M \text{ and } \tau_{\text{fast}} = 1/\tau_M + 1/\tau_r \quad (6)$$

From the fittings we observed, in all of the cases, experimental τ_{slow} is well correlated to the τ_M as obtained from the theoretical prediction using the Stokes–Einstein equation ($\tau_M = \eta v/kT$). This fact justifies the high acceptability of this model in our case. The rotational diffusion constant (D_w) due to the wobbling-type motion of the fluorescent center is given by the following equation⁵¹

$$D_w = \{\tau_r(1 - \beta)\}^{-1}[-x^2(1 + x)^2\{\ln[(1 + x)/2] + (1 - x)/2\}\{2(1 - x)\}^{-1} + (1 - x) + (6 + 8x - x^2 - 12x^3 - 7x^4)/24] \quad (7)$$

where $x = \cos \theta$; θ is the semicone angle. x is obtained from the coefficient β of biexponential fitting to the experimental anisotropy decay curve using the following equation

$$\beta = \{0.5x(1 + x)\}^2 \quad (8)$$

All rotational fitting parameters obtained from the fittings using this model are summarized in Table 1. We also studied the pH

Table 1. Wobbling-In-Cone Model Parameters Obtained from the Fittings of Rotational Anisotropy Decays of Dendrimer Molecules (~ 0.2 mM Aqueous Solution) Utilizing Their Intrinsic Fluorescence

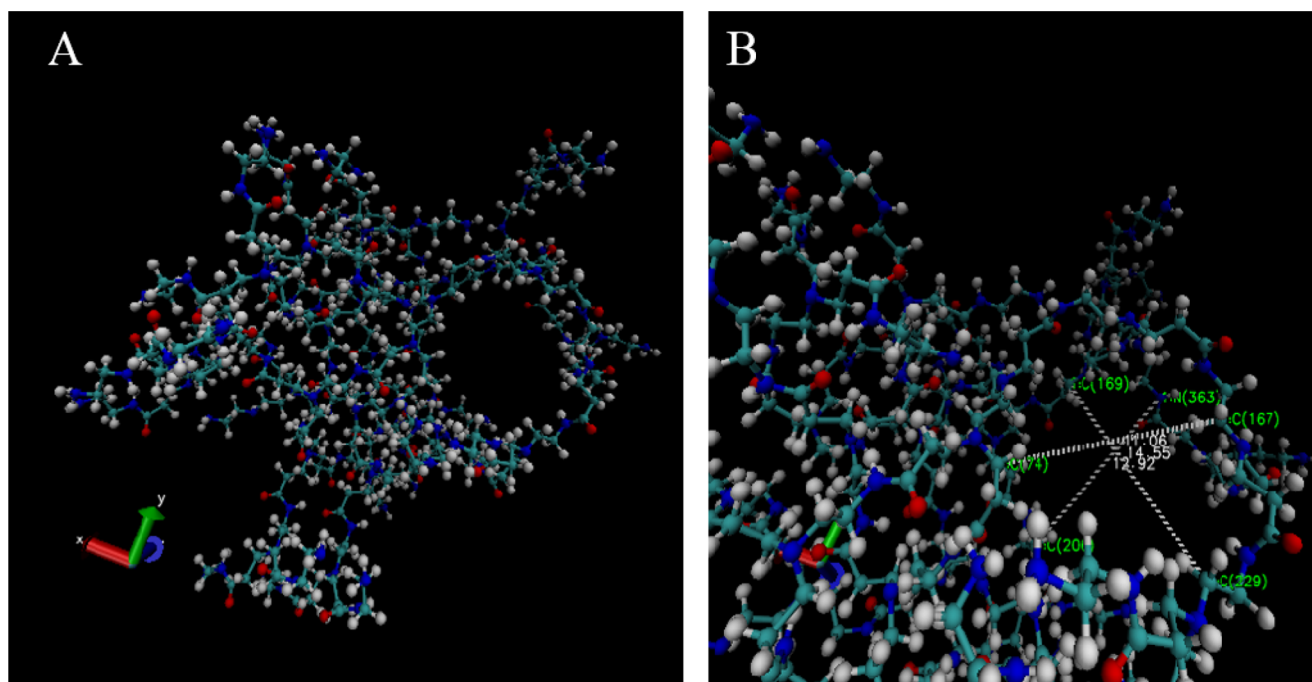
dendrimer	η (cP)	r_0	β	τ_R (ns)	τ_M (ns)	θ (deg)
G1	1.05	0.27	0.47	0.11	0.58	46
G3	1.08	0.25	0.57	1.31	3.80	30

dependency of the τ_{slow} component to get an idea of the change in dendrimer size with medium pH (τ_{slow} is directly correlated to the dendrimer size). Aqueous dendrimer solution is basic in nature (pH ~ 9.6), where we obtained $\tau_{\text{slow}} \sim 4.12$ ns for dendrimer-G3, representing an overall size of ~ 3.2 nm. However, when the pH was lowered, τ_{slow} became faster. For instance, at pH ~ 2 , $\tau_{\text{slow}} \sim 2.78$ ns, with a correlating particle size of ~ 2.8 nm (Table S2). The sizes of dendrimers obtained here through rotational anisotropy studies are highly congruent with previous reports of dendrimer sizes.^{54,55}

We also studied DLS for a size estimation of dendrimer-G3 at various pHs. DLS study reveals an insignificant change ($<5\%$) in dendrimer (G3) size with changing the medium pH from ~ 9.6 (aqueous dendrimer) to ~ 2 (aqueous dendrimer + HCl; Figure S5). The reason for the slight shrinking ($<10\%$) of dendrimer size at low pH is not clear. It might be due to the back-folding of dendrimer's peripheral subgroups toward the core at low pH. In this perspective, we may recall that Costa and his co-workers reported a similar observation of shrinking of dendrimer size at low pH.⁵⁶ Their studies of the fluorescence anisotropy decay of an anionic meso-tetrakis(4-sulfonatophenyl)porphine (TSPP) dye in an aqueous PAMAM dendrimer-G4 solution show that the rotational correlation time shrinks from ~ 9.05 to ~ 5.4 ns when pH is lowered to ~ 2 .⁵⁶ However, several other reports based on MD simulations, SANS, and the present work on DLS and rotational anisotropy show that the overall size of dendrimer scarcely changes with changing the medium pH.^{22–24} In summary, two components of the rotational anisotropy decay of dendrimer molecules may arise from two independent phenomena: (1) a complete excited-state depolarization of fluorescent center due to the overall rotation of the dendrimer and (2) a partial depolarization through the restricted wobbling-type motion of the intrinsic fluorescent center. The dimension of a dendrimer molecule in aqueous medium changes slightly with the medium pH.

3.3.2. Rotational Diffusion of C153 Attached to a Dendrimer Molecule. C153 attached to a dendrimer molecule shows a fast single-exponential rotational diffusion (~ 150 ps), which is comparable to the rotational diffusion time (~ 100 ps) of C153 in bulk water (Figure 5B).⁵⁷ Waterlike rotation of C153 attached to a dendrimer is a clear indication that C153 resides in the water-rich regions within the complex. This is congruent with our previous observations of similar bulk-water-like emission peak position (red-shifted) and bulk-water-like low quantum yield of C153 attached to a dendrimer molecule. These facts unanimously confirm that C153 is located in the most hydrophilic region of a dendrimer molecule.

Scheme 1. (A) Molecular Mechanics (MM2) Optimized Structure of PAMAM Dendrimer-G3 in the Presence of ~200 Water Molecules;³⁷ (B) Magnified Image of One Water-Rich Cavity of the Dendrimer for Inspection of the Cavity Size, Showing an Average Diameter of ~1.3 nm^a



^aC153, with a size of ~0.8 nm, can easily be accommodated within the cavity. Water molecules are not shown for clarity

We may think that waterlike environment sensed by C153 in an aqueous dendrimer solution could be due to the unbound C153 present in the solution. However, this is not the case here due to the following reasons: (1) in bulk water, saturated C153 exhibits an OD $\sim <0.02$ at its absorption peak, which increases drastically to >0.4 in the presence of ~ 0.2 mM dendrimer. As mentioned earlier, in the latter case, an intense yellow solution is observed, whereas in the former case, the resultant solution remained almost transparent (Figure S1). (2) FCS study, which will be discussed elaborately in the succeeding paragraph, shows a significant shortening of the translation diffusion time of C153 within the aqueous dendrimer solution compared to that in neat water. These facts provide direct evidence of the strong partitioning of C153 into the internal cavities of the dendrimer molecule (Scheme 1).

3.4. FCS Study of Translational Diffusion of C153–Dendrimer Complex. When C153 binds to a dendrimer molecule in an aqueous solution, it exhibits a QY of ~ 0.076 (at pH ~ 9.6 , Table 2). However, in bulk water, in the absence of

dendrimer, this value increases to ~ 0.12 .³⁸ As required by the essential conditions of FCS measurement, we used only an insignificant amount (<10 nM) of C153 with ~ 0.1 mM aqueous dendrimer-G3 solution. Such a small amount of C153 can dissolve into bulk water. Therefore, in this scenario, C153 will show a partition equilibrium between bulk water and dendrimer in the aqueous dendrimer solution. As QY is different in these two mediums, we should use the following equation for the fitting of FCS autocorrelation curve of C153 from the aqueous dendrimer solution⁵⁰

$$G(\tau) = \frac{f_W Q_W^2 D_W + f_D Q_D^2 D_D}{(f_W Q_W + f_D Q_D)^2} \quad (9)$$

where f , Q , and D represent the fraction of C153 present in water (f_W) and dendrimer (f_D); QY of C153 in water (Q_W) and dendrimer (Q_D); and translation diffusion terms (as described in ref 50) of C153 in water (D_W) and when attached to a dendrimer molecule (D_D), respectively. We used low excitation power to avoid any triplet-state kinetics within the autocorrelation curve. We found two components of translational diffusion of C153 from the aqueous dendrimer solution. The fast component ($\tau_{D1} \sim 57\text{--}62$ μs) is well correlated to the size of C153 (0.8–0.85 nm). Therefore, the source of this component is the diffusion of unbound C153 in water. The other component (τ_{D2}), which appears within the range of approximately 278–330 μs (depending on medium pH), roughly correlates to the size of dendrimer-G3 (Table 2, Figure 6). Therefore, we can assign this component because of the translational diffusion of C153 along with a dendrimer molecule. We observed here that the size of a dendrimer (G3) molecule is not much sensitive to the medium pH. The average size of the dendrimer-G3 (~ 3 nm), we obtained from our FCS study, is highly congruent with the sizes we obtained from the

Table 2. Parameters Obtained from the Fittings of FCS Autocorrelation Curves for the Translational Diffusion of C153 in Aqueous Dendrimer-G3 Solution (~ 0.1 mM) at Different pHs

pH	Q_D	τ_{D1} (μs) ^a	f_W	τ_{D2} (μs) ^a	size (nm)	f_D
3	0.11	62	0.55	278	3.14	0.45
5	0.10	57	0.45	305	3.40	0.55
7	0.092	55	0.40	282	3.20	0.60
9.6	0.076	58	0.33	330	3.66	0.67

^a τ_D is the translational diffusion time, which is related to the translational diffusion coefficient ($\tau_D = W_{xy}^2/4D$); D was obtained from the fitting of autocorrelation curve using eq 9.

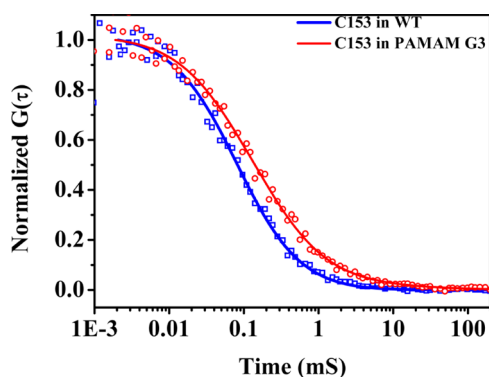


Figure 6. Normalized autocorrelation curves for C153 in water (blue) and C153 in aqueous dendrimer-G3 solution (red). pH of the aqueous dendrimer solution is ~ 9.6 .

DLS and rotational anisotropy studies. We also tried to collect autocorrelation curve using the intrinsic fluorescence of dendrimer but unable to detect it (even at low pH) due to its weak fluorescence intensity.

The fraction of C153 bound to the dendrimer (f_D) was found to increase with increasing medium pH. For instance, at pH ~ 9.6 , 67% of C153 remains attached to the dendrimer, which falls to $\sim 45\%$ at pH ~ 3 (Table 2). A similar type of observation was reported earlier by Imae and her co-worker by using a different technique.¹³ For a poly(propyleneimine)(G3)-core:PAMAM(G3)-shell dendrimer, Imae et al. observed that the number of encapsulated pyrene molecules per dendrimer increases with increasing medium pH and the value of this ratio is maximum (~ 2.7) at pH ~ 11 . The autocorrelation curve exhibits an initial fast decay (in microseconds) because of an intensity fluctuation arising from binding/unbinding dynamics between the host and guest molecules for a weak host–guest complex.^{58–60} This fast dynamics is succeeded by a slow decay (in milliseconds) in the correlation curve later because of the translational diffusion of the guest fluorescent molecule.^{58–60} Generally, these two fundamentally different processes take place in two different time regimes without interfering each other within the autocorrelation curve. In this study, we did not note any binding/unbinding dynamics of the C153–dendrimer complex in the autocorrelation curve initially.

3.5. Interaction of Dendrimer with a Hydrophilic Dye C343. The interaction of a hydrophilic fluorescent dye, C343, with dendrimer was found to be completely different from the interaction with a hydrophobic dye, C153. Unlike C153, C343 is highly soluble in water and bears a $-\text{COOH}$ functional group with a $\text{p}K_a$ value of ~ 4.65 .^{61–63} At a sufficiently low pH (< 3), all of the amine centers of dendrimer turn protonated (Scheme S2), and C343 also remains protonated at this pH. Therefore, at a low pH, the neutral form (protonated) of C343 weakly interacts with a positively charged (protonated) dendrimer. The average rotational correlation time of C343 in the aqueous dendrimer-G3 was found to be ~ 200 ps at pH ~ 2 (Figure 7), and the rotational correlation time of C343 in bulk water is ~ 140 ps.⁶⁴ This clearly indicates that most of the C343 molecules reside within the bulk water at this pH. At a high pH (~ 9), only the terminal primary amine centers of the dendrimer remain protonated (positively charged), and C343 mostly remains in its deprotonated form (negatively charged). A strong electrostatic interaction between the oppositely charged C343 and dendrimer leads to the formation of a complex, which causes a slow rotation of C343 around pH ~ 9 .

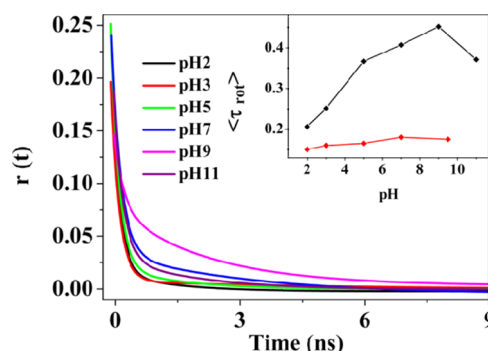


Figure 7. Fitted lines of the rotational correlation function $[r(t)]$ of C343 in ~ 0.2 mM aqueous dendrimer solution at different medium pHs (the raw data along with these fitted curves are provided in Figure S6). The inset depicts the average rotational time of C343 (black points) and C153 (red points) vs medium pH. The samples were excited at 440 nm, and the emission data were collected at the emission peak positions.

We observed almost approximately two times retardation (Figure 7) of the rotational correlation time of C343 in the aqueous dendrimer solution when pH was increased from ~ 2 ($\tau_{\text{rot}} \sim 200$ ps) to ~ 9 ($\tau_{\text{rot}} \sim 450$ ps). On further increasing the pH to 11, we observed that the rotational correlation time of C343 again becomes faster, reaching ~ 370 ps (Figure 7). At a very high pH (≥ 11), all amine centers of the dendrimer turn deprotonated ($\text{p}K_a$ of tertiary and primary amines are ~ 6.3 and 9.2 , respectively), and C343 remains deprotonated at this pH too (negatively charged). Therefore, at a very high pH, negatively charged C343 does not interact much with a neutral dendrimer molecule, which results in a recovery of the fast rotational time of C343 within the highly basic dendrimer solution. When the same experiment was conducted using C153 instead of C343, we hardly found any change in the rotational correlation time with changing the medium pH (red points in the inset of Figure 7). In all pHs, C153 exhibits a fast rotational time similar to that of C153 in bulk water (Figure 7). This is a clear indication of C153 residing within the water-filled cavities inside the dendrimer and interacting equally with the dendrimer in all pHs. For C153, we could not increase the pH beyond 10, as the spectral shapes of the dye become highly distorted at very high pHs of the medium.

4. CONCLUSIONS

In this study, using the spectral properties of C153 when attached to a dendrimer molecule, we showed that bulk-water-like environment around the C153 is highly preserved within the C153–dendrimer complex. Several spectroscopic studies (emission spectra, fluorescence anisotropy, and FCS) have substantiated that the location of C153 in the C153–dendrimer complex is most likely within the water-rich cavities inside the dendrimer molecule. This is a remarkable observation, as C153 is well known for its strong hydrophobic nature. However, a large hydrophilic molecule (bengal rose) can easily get trapped within the water-rich cavity of the dendrimer molecule.^{47–49} Our molecular mechanics (MM2)-optimized structure of dendrimer-G3 (in water) shows that the average diameter of a water-rich cavity inside dendrimer-G3 is ~ 1.3 nm, which can easily accommodate one C153 molecule with a size of ~ 0.8 nm (Scheme 1).⁶⁵ We have highlighted the most promising property of the dendrimer molecule: the formation of a stable host–guest complex with a hydrophobic molecule, and the

solubilization of the hydrophobic molecule in aqueous phase. For instance, the almost water-insoluble dye C153 is found to be greatly soluble in the presence of ~ 0.2 mM dendrimer-G3. We may be surprised over the fact that how a hydrophobic dye is largely trapped within the water-rich cavities inside a dendrimer molecule? In this perspective, it may be recalled that in a recent work Sahu and his co-worker observed a strong partitioning of water-insoluble *n*-octanol into a tightly confined water droplet formed by a reverse micelle.²⁵ However, when they increased the droplet size from ~ 3.6 to ~ 6.2 nm, *n*-octanol again became insoluble.²⁵ Sarkar and his co-workers also observed a similar phenomenon: a hydrophobic dye, C153, can be easily dissolved in a small water pool of a reverse micelle (*n*-heptane/AOT/water) but not in bulk water.⁶⁶ The water-rich peripheral cavity of the dendrimer here is even smaller than the dimensions of the water droplets of an AOT reverse micelle; therefore, the partitioning of C153 into the dendrimer cavity is not odd in our case.

■ ASSOCIATED CONTENT

§ Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.jpcb.6b10176.

Photograph of cuvettes under UV illumination, REES for both generations of dendrimer, absorption spectra of C153 in water in the presence and absence of dendrimer, dendrimer-G3 size measurement using DLS, rotational anisotropy decay of C343 in aqueous dendrimer solution at different pHs, table for anisotropy decay parameters, and schematic representation of the dendrimer structures (PDF)

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Notes

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■ ACKNOWLEDGMENTS

S.G. thanks Ramanujan fellowship grant (SR/S2/RJN-36/2012), DST, India, for supporting this study. The authors sincerely thank Prof. Kankan Bhattacharyya, Indian Association for the Cultivation of Science, Kolkata, for allowing them to use the confocal microscope.

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